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Multimodal Neuroimaging of Fronto-limbic Structure and Function Associated with Suicide Attempts in Adolescents and Young Adults with Bipolar Disorder

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Abstract

Objective—Bipolar disorder is associated with high risk for suicide behavior that often develops in adolescence/young adulthood. Elucidation of involved neural systems is critical for prevention. This study of adolescents/young adults with bipolar disorder with and without history of suicide attempts combines structural, diffusion tensor and functional magnetic resonance imaging methods to investigate implicated abnormalities in structural and functional connectivity within fronto-limbic systems.

Method—Participants with bipolar disorder included 26 with a prior suicide attempt and 42 without attempts. Regional gray matter volume, white matter integrity and functional connectivity during processing of emotional stimuli were compared between groups and differences were explored for relationships between imaging modalities and associations with suicide-related symptoms and behaviors.

Results—Compared to the non-attempter group, the attempter group showed reductions in gray matter volume in orbitofrontal cortex, hippocampus and cerebellum; white matter integrity in uncinate fasciculus, ventral frontal and right cerebellum regions; and amygdala functional connectivity to left ventral and right rostral prefrontal cortex ($p < 0.05$, corrected). In exploratory analyses, among attempters, right rostral prefrontal connectivity was negatively correlated with suicidal ideation ($p < 0.05$), and left ventral prefrontal connectivity was negatively correlated with attempt lethality ($p < 0.05$).

Conclusions—Adolescent/young adult suicide attempters with bipolar disorder demonstrate less gray matter volume and decreased structural and functional connectivity in a ventral fronto-limbic

neural system subserving emotion regulation. Among suicide attempters, reductions in amygdala-prefrontal functional connectivity may be associated with severity of suicide ideation and attempt lethality.

Introduction

As many as 56% of individuals with bipolar disorder attempt suicide and 15–19% die by suicide (1). Suicidal behavior is a leading cause of death and often commences during adolescence/young adulthood, when fronto-limbic systems implicated in suicide behavior are maturing. Thus, elucidating neural underpinnings of suicide behavior during adolescence/young adulthood may improve early identification and prevention.

Convergent neuroimaging evidence in adult attempters across psychiatric disorders supports involvement of fronto-limbic system structural and functional abnormalities. Structural magnetic resonance imaging (sMRI) and functional neuroimaging [functional MRI (fMRI) and positron emission tomography] studies in adult attempters have shown decreased gray matter volume and frontal dysfunction during emotion stimulus processing (2), particularly in ventral prefrontal cortex. Postmortem findings in suicides suggest decreased ventral prefrontal neuroreceptor binding and neurotrophic factors (3, 4). Also implicated are ventral prefrontal connection sites subserving emotion and impulse processing/regulation, including anterior cingulate cortex, insula, thalamus, striatum, and amygdala (5).

Suicide behavior has been linked to reduced structural integrity and functional fronto-limbic connections in adults in postmortem (6), sMRI, diffusion tensor imaging (DTI), and fMRI studies (5), though few were conducted in bipolar disorder specifically.

Data on neural circuitry associated with suicide attempts in adolescents/young adults are scant. Initial reports suggest consistencies with adults in reduced frontal gray matter, altered frontotemporal system responses to negative emotional stimuli and white matter abnormalities (5). Adolescent sample sizes have been low and most have had major depressive disorder.

Single modality structural and functional imaging studies in attempters implicate structure-function relationships, suggesting a need for multimodal studies. Few studies have examined associations of imaging measures with suicide-related symptom/behaviors, such as attempt lethality, ideation, intent, impulsivity and hopelessness. In adult attempters, ventral prefrontal volume decreases and dysfunction have been linked to lethality (7, 8), frontal dysfunction and white matter abnormalities to ideation (9, 10), and orbitofrontal white matter fractional anisotropy decreases to impulsivity in bipolar disorder (11). Imaging studies of suicide ideation are rare (1); one study of ideation in pediatric epilepsy showed decreased orbitofrontal white matter (12).

This study in bipolar disorder of adolescent/young adult attempters, compared to non-attempters, combines sMRI, DTI and fMRI connectivity methods to study gray matter volume, white matter integrity and functional connectivity. We hypothesized attempters would show lower ventral frontal and mesial temporal gray matter, lower white matter fractional anisotropy and functional connectivity in their connections. We explored

relationships between imaging findings across modalities and with attempt lethality, anticipating ventral prefrontal associations, and with suicidal ideation and intent, hopelessness and impulsivity. We explored comparisons to a healthy adolescent/young adult group.

Method

Subjects

Participants included 68 with bipolar disorder (age range: 14–25yrs): 26 with ≥ 1 suicide attempt (attempters), 42 without attempts (non-attempters) (Table 1). A healthy comparator group ($n=45$, $M_{age}=20.8\pm 3.3$; range 15–25, 58% female) had no lifetime DSM-IV Axis I disorders or suicide attempt or first-degree relative with a major mood or psychotic disorder or suicide attempt. Family history was assessed with the Family History-Research Diagnostic Criteria (13). Written informed consent was obtained from participants ≥ 18 yrs, and permission from parents/guardians and assent from minors, in accordance with the Yale University Institutional Review Board. Presence/absence of Axis I diagnoses and mood state at scanning were confirmed with the Structured Clinical Interview for DSM-IV Diagnosis (First MB, Spitzer RL, Gibbon M, Williams JBW, NY, 1998) for participants ≥ 18 yrs and Kiddie-Schedule for Affective Disorders and Schizophrenia (14) if < 18 yrs. Adolescents were mid-to-late pubertal. No subject had history of loss of consciousness ≥ 5 min, or neurological/major medical disorder, except treated hypothyroidism [2(8%) attempters, 1(2%) non-attempters].

Suicide attempts, defined as self-injurious acts committed with at least some intent to die, were assessed using the Columbia Suicide History Form (15) and medical lethality with the Beck Medical Lethality Scale (16). The Beck Scale for Suicide Ideation (17) assessed suicide ideation severity and Suicide Intent Scale (16) past intent to die. Additional assessments included the Beck Hopelessness Scale (18) and Barratt Impulsiveness Scale-11 (19).

MRI Acquisition

The scanning session for each subject included high-resolution sMRI, DTI and fMRI using a single 3-Tesla Siemens Trio MR scanner (Siemens, Erlangen, Germany). Sagittal sMRI images were acquired with a three-dimensional magnetization-prepared rapid acquisition gradient-echo T_1 -weighted sequence with parameters: repetition time (TR)=1500ms, echo time (TE)=2.83ms, matrix=256 $\times$ 256, field of view=256mm $\times$ 256mm, 160 1mm slices without gap and two averages. DTI acquisition was aligned with the anterior commissure-posterior commissure plane. Diffusion sensitizing gradients were applied along 32 non-collinear directions with b-value=1000sec/mm² and an acquisition without diffusion weighting b-value=0, TR=7400msec, TE=115msec, matrix=128 $\times$ 128, field of view=256mm $\times$ 256mm and 40 3mm slices without gap. FMRI data were acquired with a single-shot echo-planar imaging sequence aligned to DTI: TR=2000msec, TE=25msec, matrix=64 $\times$ 64, field of view=240mm $\times$ 240mm and 32 3mm slices without gap. Participants performed an fMRI event-related emotional face task (20), viewing faces depicting happy, neutral or fearful expressions with button press to indicate if a female or male face.

SMRI Processing

Images were processed with Statistical Parametric Mapping 5 (SPM5) (<http://www.fil.ion.ucl.ac.uk/spm>), with the segmentation function implemented for bias correction, spatial normalization and segmentation of the original structural images in the same model. SPM5 tissue probability maps (voxels $2 \times 2 \times 2 \text{mm}^3$) guided normalization and segmentation. To ensure overall tissue amount in a class was not altered, a “modulation” step was used during normalization. Gray matter images were smoothed with an 8mm full-width-at-half-maximum (FWHM) isotropic kernel (21).

DTI Processing

Diffusion-weighted data were interpolated to 1.5mm isotropic voxels and denoised by a three-dimensional isotropic Sigma 2mm FWHM Gaussian kernel. Diffusion eigenvectors and corresponding eigenvalues ($\lambda_1, \lambda_2, \lambda_3$) were acquired after diagonalization of DTI data. Fractional anisotropy was calculated and whole brain fractional anisotropy maps were normalized with SPM5 to Montreal Neurological Institute (MNI) space using a tissue probability map of white matter template, resampled to $2 \times 2 \times 2 \text{mm}^3$ during normalization and spatially smoothed at 10mm FWHM (22).

Functional Connectivity Processing

FMRI processing was performed with SPM8 (20). The first two images of each run were discarded to account for hemodynamic delay. Remaining images were realigned, then spatially normalized to MNI space. FMRI data were resampled to voxels $3 \times 3 \times 3 \text{mm}^3$ during normalization and smoothed at 8mm FWHM. Event-related response amplitudes were estimated using the general linear model (23) at the individual subject level for each of the three emotion event types: happy, neutral and fearful expressions. For each subject, this created statistical images of the BOLD signal change related to the three emotional event types versus the baseline fixation cross-hair control (24). A bilateral amygdala seed region of interest (ROI) was defined with the Wake Forest University PickAtlas Tool (<http://www.fmri.wfubmc.edu/download.htm>). The amygdala was chosen because it is implicated in suicidal behavior (5), has strong connectivity with other implicated regions showing decreased connectivity to it in adolescents/adults with bipolar disorder (20), and it may be less likely than cortex to show age-related changes in health and bipolar disorder (25, 26). For each subject, a mean time series for amygdala for each emotion type (happy, neutral, fearful) was calculated by averaging the time series for all ROI voxels. Correlational analyses were performed between the time series for the bilateral amygdala ROI with that for each brain voxel, resulting in a correlation map including the correlation coefficient for each voxel with that of the amygdala ROI, for each emotion type (20, 22). Correlation coefficients were transformed to Z-values using Fisher r-to-z transformation.

Statistical Analyses

Potential differences in age and gender across all three groups were assessed by one-way analysis of variance (ANOVA) and chi-square tests, respectively. Within the bipolar sample, two-sample *t*-tests and chi-square tests were utilized to assess potential attempter group differences in impulsivity, hopelessness, suicide ideation severity, medication status (on/off),

lifetime comorbid substance abuse/dependence (yes/no), mood state (euthymic, depressed, elevated), psychosis and rapid-cycling.

Within Imaging Modality Analyses—For primary hypothesis testing, two-sample *t*-tests were conducted in SPM to assess group differences in gray matter volume and fractional anisotropy, with age as a covariate. Findings were considered significant at $p < 0.005$ (uncorrected), if also meeting a spatial extent empirically determined by Monte Carlo simulation implemented in AlphaSim for a corrected significance of $p < 0.05$. Group differences in functional connectivity were analyzed using two-sample *t*-tests in SPM, with age as a covariate and functional connectivity correlation coefficients (Z-scores) from amygdala to brain voxels as the dependent variables for each face condition (happy, neutral, fearful). Findings were Bonferroni corrected for the three emotion conditions at $p < 0.0016$, and spatial extent empirically determined by Monte Carlo simulation. Results that survived to $p < 0.001$, with and without AlphaSim spatial extent thresholding ($p < 0.05$, corrected), are noted. Gray matter volumes, fractional anisotropy and functional connectivity correlation coefficients (Z-scores) were extracted from each cluster in which group differences were detected (MarsBar toolbox, <http://marsbar.sourceforge.net>).

Gender and medication effects on group differences were assessed with ANCOVA, but were not significant and dropped for parsimony. Exploratory analyses employed two-tailed two-sample *t*-tests to determine whether extracted values in regions of difference differed by clinical factors: medication status, lifetime substance use disorders, psychosis or rapid-cycling, and by one-way analysis of variance for mood state.

Associations between Imaging Modalities and with Behaviors—Correlational analyses were performed between gray matter volumes, fractional anisotropy, and functional connectivity correlation coefficients (Z-scores) from each cluster in which group differences were detected. Relationships between gray matter, white matter and functional connectivity abnormalities with suicide-related symptoms/behaviors (maximum lethality of attempts, intent for most lethal attempt, most severe ideation, and current hopelessness and impulsivity) were assessed using Spearman (r_s) or Pearson (r) correlations as appropriate. Correlation analyses were performed in attempters only, and for all subjects, partial correlations (r_{adj}) were estimated controlling for group (attempter vs. non-attempter) using analysis of covariance (ANCOVA). Given the exploratory nature of analyses, correlations were considered significant at $p < 0.05$, uncorrected.

Healthy Group Comparisons—In secondary analyses, gray matter volumes, fractional anisotropy and functional connectivity correlation coefficients (Z-scores) were extracted for the healthy comparison group in each region of attempter and non-attempter group differences. Three-group (healthy comparison, attempter, non-attempter) one-way ANOVAs were performed, followed by Dunnett's *t*-tests to control for multiple comparisons.

Results

Attempters and non-attempters did not differ from each other or from healthy controls in age or gender. Attempters and non-attempters differed significantly in suicide ideation severity

[$t(60) = 5.95$, $p < 0.001$], but not impulsivity, hopelessness, medication status, lifetime comorbid substance abuse/dependence, mood state, psychosis or rapid-cycling.

Group Differences Within Imaging Modalities

SMRI—Attempters demonstrated significantly lower gray matter volume than non-attempters in right orbitofrontal cortex and hippocampus, and bilaterally in cerebellum extending into the vermis (Table 2, Figure 1). Results for all regions survived to $p < 0.001$ and right orbitofrontal and hippocampus survived AlphaSim spatial extent thresholding ($p < 0.05$, corrected). No areas showed significantly increased gray matter.

Results were unrelated to clinical factors explored.

DTI—Attempters demonstrated significantly lower fractional anisotropy than non-attempters in a left uncinate fasciculus region extending into left ventral frontal areas, a right uncinate region, and right cerebellum (Table 2, Figure 2). Results for all regions survived to $p < 0.001$ and for right uncinate and right cerebellum survived AlphaSim spatial extent thresholding ($p < 0.05$, corrected). No areas showed significantly increased fractional anisotropy.

Results were unrelated to clinical factors, except fractional anisotropy in the right uncinate region was lower in bipolar disorder participants with, compared to without, rapid-cycling [$t(65) = 2.07$, $p = 0.042$].

Functional Connectivity—Attempters demonstrated significantly lower functional connectivity than non-attempters from amygdala to a left ventral prefrontal region, including orbitofrontal, rostral prefrontal and ventral anterior cingulate regions in happy and neutral conditions (Table 2, Figure 3). A similar region showed lower connectivity in the fearful condition; however, it was below the AlphaSim spatial extent threshold. There was a region of significantly lower connectivity in right rostral prefrontal cortex in the neutral condition. Significance for all regions survived to $p < 0.001$, and for all regions but the left ventral prefrontal region in the fear condition survived AlphaSim spatial extent thresholding ($p < 0.05$, corrected). No areas showed significantly increased functional connectivity.

Results were unrelated to clinical factors explored.

Associations between Imaging Modalities and with Behaviors

Bilateral cerebellum gray matter volume was significantly correlated with right uncinate fasciculus fractional anisotropy among attempters only ($r = 0.53$, $p = 0.008$), and with right and left uncinate among all subjects ($r_{\text{adj}} = 0.32$, $p = 0.009$, $r_{\text{adj}} = 0.28$, $p = 0.027$). Among all subjects, right orbitofrontal and bilateral cerebellum gray matter volumes were significantly correlated with right cerebellum fractional anisotropy ($r_{\text{adj}} = 0.25$, $p = 0.047$, $r_{\text{adj}} = 0.29$, $p = 0.022$).

Among attempters, suicide ideation severity was negatively correlated with functional connectivity in the right rostral prefrontal cortex ($r_s = -0.490$, $p = 0.033$). This correlation was not significant in the overall sample adjusted for group. Attempt lethality was negatively

correlated with left ventral prefrontal functional connectivity during happy, neutral, and fear conditions ($r_s = -0.47$, $p = 0.031$, $r_s = -0.49$, $p = 0.025$, $r_s = -0.58$, $p = 0.006$).

Healthy Group Comparisons

SMRI—Right hippocampal volume in the attempter group, but not the non-attempter group, was significantly lower than in the healthy comparison group ($p_{\text{Dunnett}} < 0.05$).

DTI—Fractional anisotropy in the attempter group, but not the non-attempter group, was significantly lower than in the healthy comparison group in the bilateral uncinate fasciculus regions ($p_{\text{Dunnett}} < 0.05$), with a trend ($p_{\text{Dunnett}} = 0.06$) in the right cerebellum region.

Functional Connectivity—Secondary analyses indicated that functional connectivity in the attempter group, but not the non-attempter group, was significantly lower than in the healthy comparison group in the left ventral prefrontal cortex region for the three face conditions ($p_{\text{Dunnett}} < 0.05$) with a trend ($p_{\text{Dunnett}} = 0.07$) in the right rostral prefrontal cortex region in the neutral condition.

Discussion

This study demonstrated reductions in gray matter volume and integrity of structural and functional connections in a fronto-limbic system in attempters, compared to non-attempters. Decreases were observed in gray matter volumes in right orbitofrontal cortex and hippocampus, and in bilateral cerebellum regions. Diminished integrity of white matter in uncinate and ventral frontal and right cerebellum regions, and decreased functional connectivity between amygdala and left ventral prefrontal and right rostral prefrontal regions were also observed. Cerebellar gray matter volume was associated with uncinate and cerebellar fractional anisotropy, and orbitofrontal gray matter volume was associated with cerebellar fractional anisotropy. Among attempters, decreases in rostral and ventral prefrontal functional connectivity were associated with severity of suicide ideation and attempt lethality, respectively.

Gray matter volume reductions associated with suicide attempts in bipolar disorder were in orbitofrontal cortex, hippocampus and cerebellum, regions implicated in emotional processing and regulation and in bipolar disorder (27). The orbitofrontal cortex plays a central role in regulating emotional responses, including through its connections to amygdala, hippocampus, and cerebellum (27). Volume decreases in the orbitofrontal cortex of attempters has been reported in adult attempters with major depressive disorder (28). The hippocampus may also influence emotional reactions and regulatory processes, given its role in encoding and recall of the emotional significance of events. Moreover, impaired memory processes have been associated with suicide attempts, implicating the hippocampus (29). Cerebellar connections with prefrontal cortex and amygdala suggest a role for cerebellum in emotion regulation, consistent with emotion dysregulation observed with cerebellum lesions (27). Hippocampus and cerebellum volume decreases have been observed in adult attempters (30, 31), but have been studied primarily in attempters with major depression, and some findings are inconsistent (28, 32). We did not detect differences in amygdala volume

between the bipolar disorder attempters and non-attempters. Varying findings may be due to differences in imaging methodology and subject samples.

White matter findings were in regions of bilateral uncinate, extending into left ventral frontal regions, and in right cerebellum. The uncinate provides major ventral prefrontal cortex-amygdala connections and is therefore implicated in emotional regulation. In bipolar disorder, white matter orbitofrontal decreases have been reported and were associated with impulsivity (11). The white matter decreases observed herein in attempters included more caudal frontal and temporal areas and were not associated with impulsivity. Right uncinate fractional anisotropy values were lower in association with rapid-cycling, a suggested risk factor for suicide in bipolar disorder (33), and suggests a link to mood dysregulation. Decreased white matter volumes have been observed in cerebellum in attempters with major depression (31). The cerebellum finding in attempters with bipolar disorder suggests it may be a structural abnormality associated with attempts across these mood disorders.

Exploratory analyses also provided preliminary evidence for associations between some of the gray and white matter findings. This raises the possibility of interactions between development of gray and white matter in suicide risk. This study's data do not permit addressing this possibility. However, reciprocal relationships between neurons and glia during development may contribute to concomitant gray and white matter developmental changes (34–36). It is also possible that gray-white associations are the results of other biological processes, as opposed to a direct interaction between their developmental processes. For example, they may reflect responses to common genetic or environmental factors.

Amygdala-ventral prefrontal functional connectivity is critical in emotion regulation and showed decreases during processing of happy and neutral faces, and a trend for fearful faces, in attempters compared to non-attempters. Studies suggest emotion regulation deficits in adolescent suicide attempters and adolescents may attempt suicide in an effort to dampen intolerable emotional states (37). Moreover, adolescents with higher affect dysregulation are more likely to have multiple suicide attempts (38). Previous fMRI studies have shown increased responses to negative emotional faces, specifically angry faces, in frontal circuitry in adolescents (39) similar to adults (2). Depression and suicidal ideation have been associated with altered responses to neutral face stimuli in adults, particularly with projecting a negative bias (40). Together with reductions in amygdala-ventral prefrontal functional connectivity in the happy condition and at the trend level in the fearful condition, this suggests more widespread affective dysregulation may be present in suicide attempters with bipolar disorder, limiting modulation of emotional responses and impairing effective management of affect.

Decreased functional connectivity from the amygdala was also noted in right rostral prefrontal cortex in the neutral condition, suggested by exploratory analyses to be associated with suicide ideation severity. Less is known about the role of the rostral prefrontal cortex in suicidal behavior. The experience of mental pain in attempters during recall of suicidal episodes showed associations with decreased functioning of the rostral prefrontal cortex (41), a region implicated in self-referential and introspective mental functions (42) altered in

suicidal ideation and during suicidal acts (43). Frontal dysfunction was observed in a study of adult ideators (9). In depressed youths, maladaptive emotion regulation response tendencies increased the likelihood of suicide ideation, planning and action, while adaptive regulatory responses decreased suicidal behavior (46). As the rostral prefrontal cortex is important in the development of regulation of emotions and adaptive responses tendencies (44), it is possible that emotional dysregulation, resulting from disruptions in prefrontal system functional connectivity, contributes to suicide ideation and risk for suicide behavior.

The magnitude of functional connectivity between amygdala and left ventral prefrontal cortex was negatively associated with attempt lethality. Individuals who make high-lethality attempts may be at most risk for suicide death (8). Our finding comports with adult neuroimaging studies demonstrating associations between attempt lethality and ventral prefrontal hypometabolism (8). High lethality has been associated with highly planned, less impulsive attempts (8). Together with the lack of associations between impulsivity and reductions in structural and functional integrity among attempters in this study, this suggests that altered ventral prefrontal functioning in attempt lethality may not, at least for some individuals, be related to its role in controlling impulses, but may relate to emotional dysregulation. However, impulsivity was assessed at scanning. It is possible associations with ventral prefrontal alterations may exist at the time of attempt.

For most regions in which attempters differed from non-attempters, attempters, but not non-attempters, also differed significantly from healthy controls. Although this suggests findings may be related to suicidal tendencies, comparisons to individuals with other disorders are needed. It is also possible findings are a consequence of bipolar disorder or attempts. Future longitudinal studies are needed to elucidate the predictive value of these results.

Orbitofrontal and cerebellum gray matter volume decreases in healthy controls compared to suicide attempters did not reach significance. This suggests gray matter volumes in these regions may better differentiate attempter status within bipolar disorder. Orbitofrontal volume decreases in attempters compared to non-attempters have been reported previously in bipolar disorder (45). Orbitofrontal decreases have also been reported between healthy comparison groups and attempter groups with major depressive disorder and borderline personality disorder (5). It is possible that aspects of our sample or methods decreased the ability to detect significant differences between attempters and healthy controls. Cerebellar findings are mixed (5) and additional studies are warranted.

Study limitations include the modest sample size. While no effects in the regions of group differences were detected for gender, medication, substance comorbidity, mood state or psychosis, the number of subjects in the subgroups for the factors assessed was small, limiting power. There were not sufficient numbers of subjects to perform meaningful analyses for subtypes of medications and non-substance psychiatric comorbidities. Age range was large and the sample included pubertal teens and young adults. While age was included as a covariate, developmental brain changes could still have affected results. Although suicide attempt and ideation history were assessed systematically, assessments were based on retrospective subject reporting, which could decrease accuracy. Results were reported with a primary threshold of $p < 0.005$, with AlphaSim spatial extent thresholding ($p < 0.05$, corrected), in order to balance effort to avoid premature strict correction that may

limit report of findings that could prove useful in the future to the field, especially as there is rare similar data in the relatively young field of suicide research, with potential report of spurious results. However, caution should be taken in considering the results, as lower primary thresholds may lead to spurious findings when utilizing cluster extent based thresholds (46). We noted which results survived to a threshold of $p < 0.001$, with and without AlphaSim spatial extent thresholding ($p < 0.05$, corrected), to provide this further information. Correlational analyses should be considered preliminary. Exploratory correlational analyses were not controlled for multiple comparisons. Although some associations across imaging modalities and with behaviors were in directions anticipated, these did not reach significance, potentially owing to the sample size. Directionality of the DTI acquisition was limited. Additional analyses (e.g. analyses of group differences in activation and/or functional connectivity analyses using psychophysiological interaction analysis methods to further isolate stimulus-related connectivity) could aid further interpretation of the data. However, as in the previous published work of our group (20, 22), and other research groups, there can be complex interactions between mood state and stimuli emotion type. Thus the sample sizes in the current study limit ability to fully assess further interactions with suicide history and additional comprehensive analyses are beyond the scope of this manuscript.

In summary, this multimodal structural and functional neuroimaging study of adolescent/young adult suicide attempters with bipolar disorder demonstrates decreases in gray matter and white matter structural and functional connectivity in a ventral fronto-limbic neural system that subserves emotion regulation. It provides preliminary evidence for gray-white matter relationships and associations between system features with suicide ideation and attempt lethality. Integrated study of brain structure and function, and their associations with symptoms and behavior, in suicide research is important to elucidating risk mechanisms that may involve complex, parallel and interacting relationships between developing circuitry and symptoms/behaviors. Longitudinal studies are needed to more fully investigate development of system features, their role in transition from suicidal ideation to behavior and risk for completed suicide, and further elucidate the role of emotional regulation in suicide behavior in bipolar disorder and whether it generalizes to other disorders.

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Biography

Dr. Oquendo receives royalties for the commercial use of the Columbia Suicide Severity Rating Scale and her family owns stock in Bristol Myers Squibb. Dr. Martin receives royalties from Wolters Kluwer and an editorial stipend from the American Academy of Child and Adolescent Psychiatry. Other authors have no financial conflicts of interest to report.

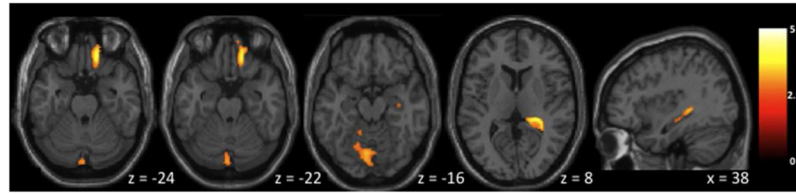


Figure 1. Decreased Gray Matter Volume in Individuals With Bipolar Disorder With, Compared to Without, Suicide Attempts^a

^a The structural magnetic resonance T1 axial-oblique and sagittal images display right orbitofrontal cortex, right hippocampal and cerebellum regions where gray matter volume was decreased in the group with bipolar disorder with suicide attempts, compared to the group without suicide attempts ($p < 0.005$, meeting AlphaSim spatial extent threshold for $p < 0.05$, corrected). The numbers are the Montreal Neurological Institute coordinates (mm) for the corresponding plane. The color bar represents the range of T -values. The right side of the axial-oblique images is on the right side of the brain.

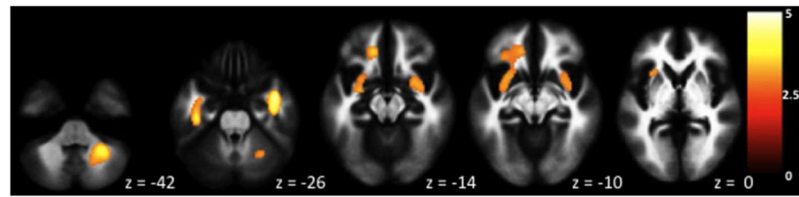


Figure 2. Decreased Fractional Anisotropy in Individuals With Bipolar Disorder With, Compared to Without, Suicide Attempts^a

^a Axial-oblique images display areas of decreased fractional anisotropy, in regions of the bilateral uncinate fasciculus, extending on the left into left ventral frontal cortex, and cerebellum in the group with bipolar disorder with suicide attempts, compared to the group without suicide attempts ($p < 0.005$, meeting AlphaSim spatial extent threshold for $p < 0.05$, corrected). The numbers are the Montreal Neurological Institute coordinates (mm) for the z-plane. The left uncinate was negatively associated with suicide ideation severity ($p < 0.05$). The color bar represents the range of T -values. The right side of the axial-oblique images is on the right side of the brain.

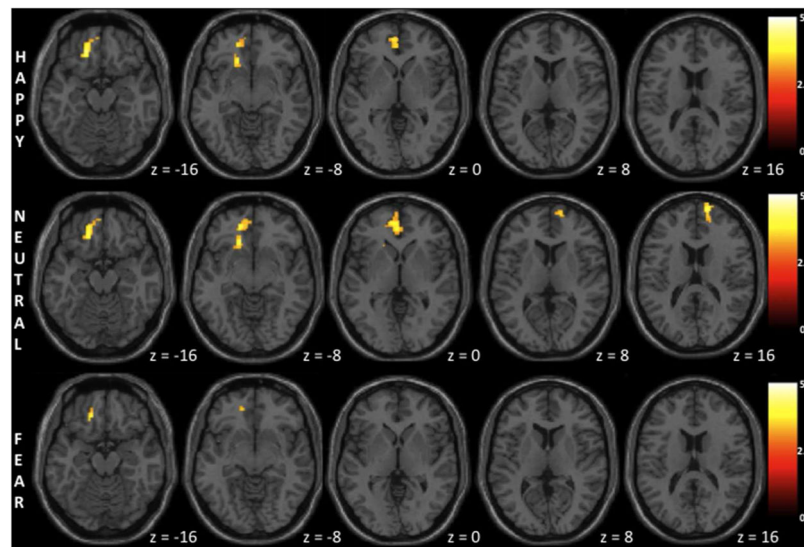


Figure 3. Decreased Functional Connectivity in Individuals With Bipolar Disorder With, Compared to Without, Suicide Attempts^a

^a Axial-oblique images display the regions of decreased functional connectivity from the amygdala, in the group with bipolar disorder with suicide attempts, compared to the group without suicide attempts, to left ventral prefrontal cortex during happy (top row, $p < 0.0016$, meeting AlphaSim spatial extent threshold for $p < 0.05$, corrected), neutral (middle row, $p < 0.0016$, meeting AlphaSim spatial extent threshold for $p < 0.05$, corrected), and fearful face (bottom row, $p < 0.0016$, uncorrected) conditions and to right rostral prefrontal cortex in the neutral face (second row, right two images, $p < 0.0016$, meeting AlphaSim spatial extent threshold for $p < 0.05$, corrected) condition. The numbers are the Montreal Neurological Institute coordinates (mm) for the z-plane. Functional connectivity in all areas was negatively associated with suicide ideation severity ($p < 0.05$). Only the region of reduced amygdala-ventral prefrontal connectivity was negatively associated with lethality of most severe attempt in all three face conditions ($p < 0.05$). The color bar represents the range of T -values. The right side of the axial-oblique images is on the right side of the brain.

TABLE 1

Demographic and Clinical Characteristics of Participants with Bipolar Disorder

Characteristic	Bipolar Disorder Participants with Attempts		Bipolar Disorder Participants without Attempts	
Age Range (years)	14–25		14–25	
	Mean	SD	Mean	SD
Age (years)	20.5	3.0	20.6	3.2
Beck Hopelessness Scale	7.3	5.6	5.8	4.4
Barratt Impulsiveness Scale-11 Total	75.1	13.2	70.2	10.0
Beck Scale for Suicide Ideation	18.1	9.7	5.1	7.4
	N	%	N	%
Female	20	77	23	55
Rapid-Cycling	14	54	13	31
Mood State at Scan				
Euthymic	11	42	23	55
Depressed	6	23	10	24
Elevated	9	35	9	21
Lifetime Psychosis	12	46	15	36
Unmedicated	8	31	16	38
Medications				
Lithium Carbonate	5	19	9	21
Anticonvulsants	10	38	9	21
Antipsychotics	9	35	21	50
Antidepressants	5	19	4	10
Stimulants	5	19	8	19
Benzodiazepines	5	19	6	14
Methadone	1	4	1	2
Levothyroxine	2	8	1	2
Adrenergic agonists	-	-	3	7
Naltrexone	1	4	-	-
Non-benzodiazepine Hypnotic	-	-	1	2
Comorbidity				
Lifetime Substance Use Disorders	8	31	15	36
Substance Abuse	4	15	4	10
Substance Dependence	5	19	5	12
Alcohol Abuse	3	12	8	19
Alcohol Dependence	2	8	-	-
Cannabis Abuse	3	12	4	10
Cannabis Dependence	3	12	3	7
Cocaine Abuse	1	4	1	2
Cocaine Dependence	1	4	2	5
Opioid Dependence	2	8	2	5
Amphetamine Abuse	1	4	-	-

Characteristic	Bipolar Disorder Participants with Attempts		Bipolar Disorder Participants without Attempts	
Polysubstance Dependence	-	-	1	2
Sedative-hypnotic Dependence	-	-	1	2
Other Substance Dependence	-	-	1	2
Lifetime Other Psychiatric Disorders				
Panic Disorder	6	23	5	12
Post Traumatic Stress	5	19	5	12
Social Phobia	2	8	3	7
Specific Phobia	2	8	1	2
Obsessive Compulsive	2	8	1	2
Generalized Anxiety	1	4	2	5
Anorexia Nervosa	1	4	1	2
Bulimia Nervosa	-	-	3	7
Eating Disorder NOS	1	4	1	2
Attention Deficit/Hyperactivity *	2	33	1	13

* Assessed only in individuals <18 years of age, N (% adolescents)

TABLE 2

Areas of Significant Difference Between Individuals With Bipolar Disorder With, Compared to Without, Suicide Attempts

Regions	Brodman Areas	Cluster Size	MNI Coordinates			T-Values
			x	y	z	
Gray Matter Volume						
Right Orbitofrontal	11/47	215	16	36	-24	p<0.005 4.16
Right Hippocampus	-	433	28	-34	8	4.23
Cerebellum	-	410	-4	-80	-18	3.35
Fractional Anisotropy						
Right Uncinate Fasciculus	-	636	42	0	-28	p<0.005 4.78
Left Uncinate/Ventral Prefrontal	-	900	-40	-22	-26	3.79
Right Cerebellum	-	579	28	-50	-42	4.06
Functional Connectivity						
Happy Face						p<0.0016
Left Ventral Prefrontal	11/47, 10, 32	109	-18	36	-15	4.31
Neutral Face						
Left Ventral Prefrontal	11/47, 10, 24/32	190	-15	27	-9	4.21
Right Rostral Prefrontal	10	71	15	63	15	4.22
Fearful Face						
Left Ventral Prefrontal	11/47, 10	20	-18	36	-18	4.11

* The table includes all regions that met AlphaSim spatial extent threshold for p<0.05, corrected, with the addition of the left ventral prefrontal region in the fearful face condition for functional connectivity that was below AlphaSim threshold.